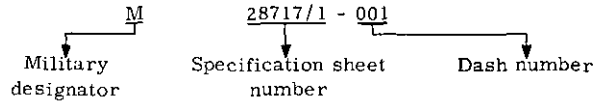
MILITARY SPECIFICATION

FILTERS, AIR, VISCOUS-IMPINGEMENT TYPE, CLEANABLE AND REUSABLE, FOR EQUIPMENT ENCLOSURES; AND IMPINGEMENT OIL GENERAL SPECIFICATION FOR

1. SCOPE

1.1 This specification covers the general requirements for air filters of the cleanable or permanent type for use in equipment enclosures requiring unit or panel type medium efficiency air filters. This specification also covers the general requirements for the impingement oils used for treating or coating of the air filter element.

1.2 Classification. The Military part number shall consist of the letter "M", the basic number of the specification sheet, and an assigned dash number (see 3.1) as shown in the following:



2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636 - Box, Fiberboard
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper (for Carton Sealing)

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-E-5272 - Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-45662 - Calibration System Requirements.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts
MIL-STD-290 - Packaging, Packing and Marking of Petroleum and Related Products
MIL-STD-810 - Environmental Test Methods
MIL-STD-889 - Dissimilar Metals.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS
D93-61 - Test for Flash Point by Pensky-Martens Closed Tester
D97-66 - Test for Cloud and Pour Points
D130-56 - Test for Copper Strip Corrosion by Petroleum Products
D445-65 - Test for Kinematic Viscosity
D568-61 - Flammability of Plastics 0.050 inch and Under in Thickness

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

AIR FILTER INSTITUTE
Code for Testing Air Cleaning Devices used in General Ventilation, Section 1, Unit or Panel Type Air Filtering Devices, 1956 Revision.

(Application for copies should be addressed to the Air Filter Institute, Inc., Box 85, Station E, Louisville, Kentucky.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Blvd., Chicago, Illinois 60606.

3. REQUIREMENTS

3.1 Specification sheets. The individual part requirements shall be as specified herein and in accordance with the applicable specification sheets. (See 6.2). In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.2 Qualification. Air filters and impingement oils furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.3 Material. Material shall be used which will enable the product to meet the requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Dissimilar metals. The use of dissimilar metals in direct contact with each other, which tend toward active electrolytic corrosion (particularly brass, copper, or steel in contact with aluminum or aluminum alloy), shall not be acceptable. Metal plating or metal spraying of dissimilar base metals to provide similar or suitable abutting surfaces shall be permitted. The use of dissimilar metals separated by suitable nonmetallic insulating material shall be permitted. Dissimilar metals are defined in Standard MIL-STD-889.

3.3.2 Filter media and frame. The filter media and frame shall be of metallic material.

3.4 Design and construction. Air filters shall be of the design, construction and dimensions specified (see 3.1). All external edges and sharp corners shall be rounded.

3.5 Viscous impingement oil. Viscous impingement oil furnished for or supplied with the filters shall be in accordance with the requirements of this specification for the properties shown (see 3.1) when tested as specified in 4.4.1.

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3.6 Temperature and thermal shock. The filters when tested in accordance with 4.6.2 (see 3.1) shall exhibit no peeling, flaking, chipping, cracking, nor crazing of the protective finish coating, no corrosion or deterioration of the base metal, no major distortion, dislocation, or discoloration of the filter element or other damage which could in any way significantly affect the Filters' rigidity and performance characteristics. All markings shall remain legible. Temperature shall be as specified (see 3.1).

3.7 Dust arrestance. When tested as specified in 4.6.6, minimum dust arrestance shall be as specified (see 3.1).

3.8 Dust-holding capacity. When tested as specified in 4.6.7, minimum dust-holding capacity shall be as specified (see 3.1).

3.9 Rated velocity. Rated velocity shall be as specified (see 3.1) and shall be used in testing performed according to 4.6.6. No damage shall result from maximum flow specified in 4.6.6.1.

3.10 Size. The filters shall conform to one of the sizes specified (see 3.1) when tested as specified in 4.6.1. The dimensions shown on the individual specification sheet refer to the overall dimensions of the filter media in its frame. Tolerances for all thicknesses are $\pm \frac{1}{32}$ inch. Tolerances for the size 20 x 20 face section are +0, $-\frac{1}{8}$ inch; all other face section tolerances are $\pm \frac{1}{16}$ inch.

3.11 Structural integrity. Filters shall satisfactorily withstand the specified vibration and shock conditions when tested according to 4.6.4. The samples shall show no evidence of permanent distortion or other damage and shall meet the requirements of 3.10 following the tests of 4.6.4.

3.12 Radio frequency attenuation. When tested as specified in 4.6.9, the radio frequency attenuation shall be as specified (see 3.1).

3.13 Cleanability. Filters shall retain at least 80 percent of the original filtering characteristics (see 3.7 and 3.8) when tested as specified in 4.6.8.

3.14 Salt spray (corrosion). Filters, when tested as specified in 4.6.5, shall show no evidence of corrosion, peeling, chipping or blistering of the finish and base metal or other damage which could in any way significantly affect the Filters' rigidity and performance. All markings shall remain legible.

3.15 Fungus resistance. The filter shall show no evidence of fungus growth when tested as specified in 4.6.3. In addition, there shall be no evidence of corrosion, peeling, chipping or blistering of the finish and base metal when inspected by the unaided eye. Damage due to decontamination shall not be cause for failure. All markings shall remain legible.

3.16 Marking. The filter shall be marked in accordance with Standard MIL-STD-130. The markings shall consist of the part number, Mfrs. name or trade symbol, date code, cleaning and retreating instructions, nominal size and an arrow indicating direction of air flow.

3.17 Weight. The weight of the Filter in grams shall not exceed 1.5 times the filter volume in cubic inches, unless otherwise specified (see 3.1).

3.18 Flammability. The filter media shall be non-burning or self-extinguishing when tested as specified in 4.6.10. In addition, the Filter shall exhibit no damage which could in any way cause disassembly of any parts.

3.19 Shock-corner drop. The Filters shall satisfactorily withstand the specified Corner Drop Shock conditions when tested according to 4.6.11. The samples shall show no evidence of permanent distortion or other damage and shall meet the requirements of 3.10 following the test of 4.6.11.

3.20 Impingement oils. The impingement oils, when tested in accordance with 4.4.1, shall meet the requirements specified (see 3.1).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Qualification inspection.
(b) Quality conformance inspection.

4.3 Standard test conditions. Unless otherwise specified herein, all measurements and tests shall be made at the following conditions:

(a) Temperature: 25° ± 10°C. (77° ± 18°F).
(b) Relative humidity: 20% to 80%.
(c) Barometric pressure: Ambient atmospheric.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government.

4.4.1 Impingement oil. The oil sample shall be subjected to the tests specified in table I. Failure to conform to the tests specified shall be cause for refusal to grant qualification.

TABLE I. Qualification tests, impingement oil.

Test	Method, ASTM Designation
Flash Point	D93-61
Pour Point	D97-66
Viscosity	D445-65
Corrosion	D130-56

4.4.1.1 Qualification sample. The qualification sample shall consist of 2 one quart units (or as close to one quart as the supplier's standard packaging permits). Each unit may consist of more than one standard package.

4.4.2 Filters. The filter sample shall be subjected to the tests specified in table II, in the order shown, except that Groups II, III, and IV may be conducted concurrently. The Filters shall be oil impregnated according to the suppliers' instructions before performance of Groups II, III, and IV. Defective specimens shall be cause for refusal to grant qualification.

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TABLE II. Qualification tests, filters

Examination	Requirement Paragraph	Test Paragraph	Units to be Tested
Group I			
Visual & mechanical	3.4, 3.10 3.16, 3.17	4.6.1	1 thru 7
Group II			
Temperature and Thermal Shock	3.6	4.6.2	1 and 2
Fungus Resistance	3.15	4.6.3	1
Structural Integrity	3.11	4.6.4	2
Group III			
Salt Spray (Corrosion)	3.14	4.6.5	3 and 4
Group IV			
Flammability	3.18	4.6.10	5
Group V			
Structural Integrity	3.11	4.6.4	6
Dust Arrestance	3.7	4.6.6	6 and 7
Dust Holding Capacity	3.8	4.6.7	6 and 7
Cleanability	3.13	4.6.8	7
Group VI			
Radio Frequency Attenuation	3.12	4.6.9	8
Shock-Corner Drop	3.19	4.6.11	8

4.4.2.1 Qualification sample. The qualification sample shall consist of eight filters of the design and construction for which qualification is desired. The filter sizes shall correspond to the face section sizes designated by Size 20 x 20 (see 3.1). Compliance to the Qualification Requirements for samples of these sizes, shall qualify samples of all other face section sizes shown (see 3.1) for the same thickness only.

4.4.2.2 Additional tests. For sizes (see 3.1) other than that submitted in the qualification sample (see 4.4.2.1), the Government may also require structural integrity and/or radio frequency attenuation tests (see 6.2). Such tests shall be at the expense of the supplier.

4.5 QUALITY CONFORMANCE INSPECTION

4.5.1 Inspection for delivery, impingement oil. Inspection of product for delivery shall consist of flash point and corrosion tests of 4.4.1.

4.5.2 Inspection for delivery, filters. Inspection of filters for delivery shall consist of Group I tests (see 4.4.2).

4.5.2.1 Sampling plan. Samples for Group I tests shall be selected in accordance with MIL-STD-105. The acceptable quality level (AQL) shall be one percent.

4.5.3 Inspection lot. An inspection lot shall consist of all units produced from one batch of material. If the material is not separable into batches, a lot shall be all units produced using the same production and inspection processes and offered for inspection at one time.

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4.5.3.1 Batch. A batch is a homogeneous mixture of materials, such that one portion of the mixture will have the same characteristics as any other portion of the mixture.

4.5.4 Additional tests. Prior to acceptance, the Government may require that representative samples of the inspection lot be submitted for any of the tests in 4.4.2. Such samples shall be selected according to MIL-STD-105. Additional tests, if required, will be at the expense of the Government. The samples for such tests shall be considered delivered with the inspection lot from which they were selected. The acceptable quality level (AQL) shall be one percent.

4.5.5 Defectives. Defective samples shall be cause for refusal of acceptance of the lot represented by the sample.

4.6 Test procedures.

4.6.1 Visual and mechanical examination. The filter shall be examined to determine conformance to this specification and the applicable specification sheet with respect to materials, workmanship, dimensions, markings, and weight.

4.6.2 Temperature and thermal shock (see 3.6). Filters shall be subjected to the high temperature test, Procedure II, of MIL-E-5272, except that the temperature shall be the highest temperature required by the filter service temperature class (see 3.1). Within 5 minutes of the conclusion of the high temperature exposure, the filter shall be exposed to the lowest temperature required by the filter service temperature class (see 3.1). This temperature shall be maintained for two hours. At the end of the low temperature exposure the filter shall be returned to room conditions and examined for conformance to 3.6.

4.6.3 Fungus resistance. Filters shall be tested as specified for Procedure I of MIL-E-5272. After the exposure the filters shall be examined for conformance to 3.15 and then decontaminated.

4.6.4 Structural integrity. Filters shall be subjected to Vibration and Shock Tests in accordance with 4.6.4.1 and 4.6.4.2.

4.6.4.1 Vibration. Filters shall be tested for vibration at room temperature conditions in accordance with Method 514 of Standard MIL-STD-810 for Procedure I (Parts 1, 2 and 3), Curve D of Figure 514-1 and Time Schedule I of Table 514-II. Upon completion of the test, the Filters shall be examined for conformance to 3.11.

4.6.4.2 Shock, medium impact. Filters shall be tested for Shock in accordance with Method 205 of MIL-STD-202 for Test Condition B. Upon completion of the test, the Filters shall be examined for conformance to 3.11.

4.6.5 Salt spray. Filters shall be tested according to Method 101, Test Condition B, of MIL-STD-202, using a 5% salt solution. The filters shall be examined for conformance to 3.14.

4.6.6 Dust arrestance. Filters shall be tested according to the Air Filter Institute Code for Testing Air Cleaning Devices for General Ventilation, Section 1, Unit or Panel Type Air Filtering Devices, 1956 revision as modified by 4.6.6.1. The arrestance of the Filter shall be as specified in 3.7.

4.6.6.1 Air flow velocity. Air velocity during the test shall be as required by 3.9. At the end of the test, flow at the maximum for the filter's class (see 3.1) shall be required for five minutes.

4.6.7 Dust holding capacity. Dust holding capacity shall be determined concurrently with dust arrestance (4.6.6). The capacity of the filter shall be as specified in 3.7.

4.6.8 Cleanability. Filters tested according to 4.6.6 and 4.6.7 shall be cleaned and retreated according to the Mfr's. specified instructions. The tests of 4.6.6 and 4.6.7 shall then be repeated to determine conformance to 3.13.

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4.6.9 Radio frequency attenuation. The Filters shall be subjected to measurements for Radio Frequency Attenuation in accordance with the general theory and practice associated with Electromagnetic Waves in hollow waveguides for Transverse-Electric Waves.

4.6.9.1 The test chamber and test set-up to be employed shall be as shown on Figure 1. The chamber shall be fabricated out of aluminum alloy material and shall consist of two identical separate sections in order to provide easy insertion of the test filter. The overall internal dimensions of the assembled chamber, without the test filter inserted, shall be approximately 18-1/4 inches length and width by 11-7/8 inches deep. The coupling rods for electromagnetic wave propagation shall be installed in both chamber sections. The rods shall be fabricated out of brass material and shall be 3/32 inches in diameter. Electrical connections from the rods to the externally located test instrumentation shall be made by means of appropriate shielded cables.

4.6.9.2 With the connections made as shown in Figure 1, measurements shall be performed at test frequencies of 1, 10, 100 and 1000 megacycles. At each frequency, the Filter Attenuation shall be determined as the difference in attenuation readings with the Filter positioned in and then removed from the test chamber. For both readings, the signal generator shall be adjusted for the same level indication on the field strength meter.

4.6.9.3 The measurements shall be repeated with the Test Filter rotated 90 degrees physically from the original position. The minimum readings obtained among both determinations shall be in accordance with the requirements specified (see 3.1).

4.6.10 Flammability. Flammability shall be determined according to ASTM Designation D568-61, except that conditioning of the specimen is not required. Thickness of the specimen shall be the full thickness of the filter. Metal, glass, and other obviously nonflammable material need not be tested. The filters shall be inspected to determine conformance with 3.18.

4.6.11 Shock-corner drop. Each Filter shall be subjected to one free full drop on each of its four corners. The Filter shall be held by one corner and allowed to drop on the opposite corner. The impacting corner shall fall 36 inches on a concrete surface. Upon completion of the test, the Filters shall be examined for conformance to 3.11.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Filters shall be packaged five each in accordance with Method III of MIL-P-116.

5.1.2 Level C. Filters shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Filters packaged as specified in 5.1 shall be packed in containers conforming to PPP-B-636, grade V2 or V3, special requirements, type 2 load, or PPP-B-636. Containers shall be sealed with tape at least $1\frac{1}{2}$ inches wide conforming to PPP-T-76 and reinforced with nonmetallic tape bonding.

5.2.2 Level B. Filters packaged as specified in 5.1 shall be packed in containers conforming to PPP-B-636 class domestic, special requirements. Closure and reinforcement shall be in accordance with the box specification. Nonmetallic tape bonding shall be used for reinforcement.

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5.2.3 Level C. Filters packaged as specified in 5.1 shall be packed in containers in accordance with uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.2.4 Impingement oil. Impingement oil shall be packed in accordance with the appropriate level (see 6.2) of MIL-STD-290 for miscellaneous oils in 5-gallon size.

5.3 Marking.

5.3.1 Filters. In addition to any special marking required by the contract or order, unit and intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Impingement oil. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-290. The minimum flash point of the appropriate class (see 3.20) shall be applied under special marking requirements.

6. NOTES

6.1 Intended use. Filters and impingement oil covered by this specification are for use in enclosures for equipment requiring forced air circulation for cooling purposes and moderate protection from air borne particulate matter.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet, and the complete part number (see 1.2 and 3.1).
(c) Levels of preservation, packaging and packing, and applicable marking (see section 5).
(d) Additional tests required (see 4.5.4).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Electronic Systems Command, Department of the Navy, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

Preparing activity:
Navy - EC

Project 4310-N033

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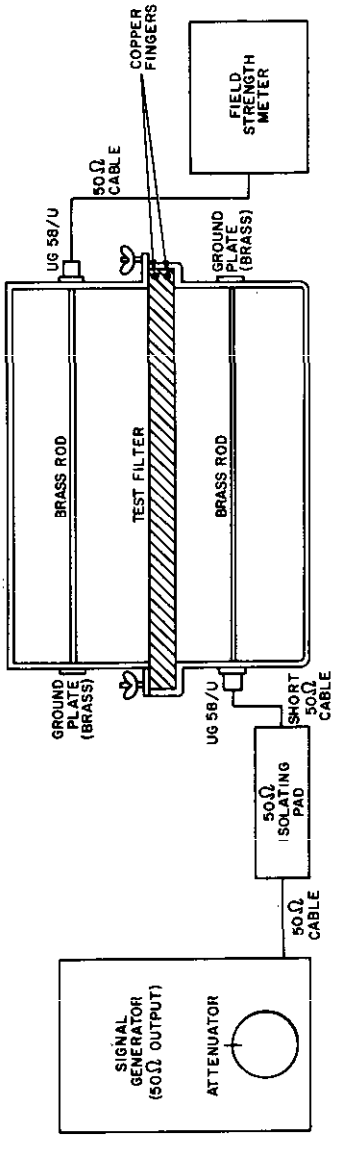


FIGURE 1 - R. F. Attenuation Test Set-up

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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